

CryorApply - System Design Document

AI-powered CV Reviewer for students and beginner job seekers

1. Introduction

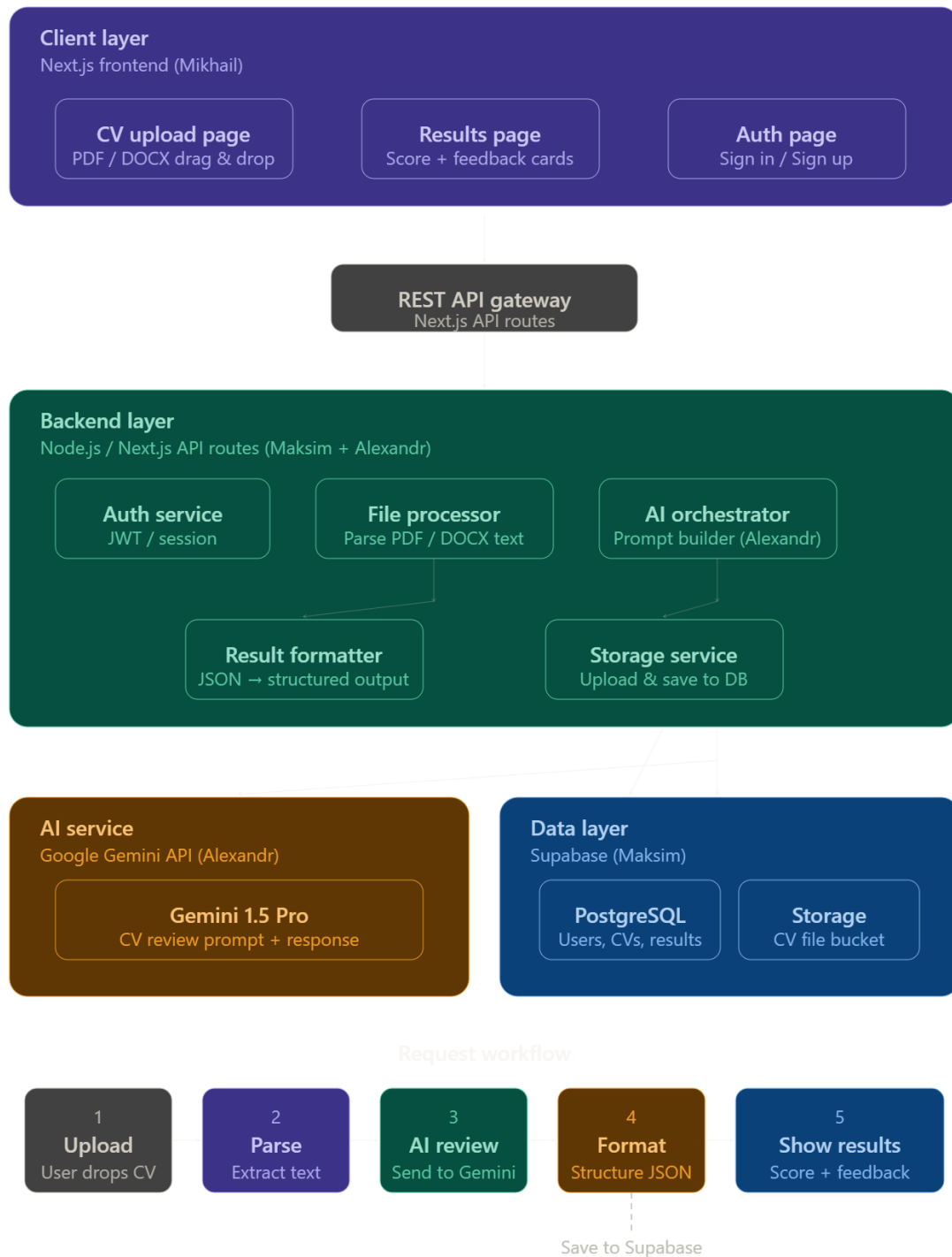
CryorApply is a web application that helps students and beginner job seekers improve their CVs before applying for jobs or internships. The platform accepts a CV file (PDF or DOCX), analyses it using the Google Gemini 3.1 Flash API, and returns structured feedback including a quality score and actionable improvement suggestions.

2. Team and Roles

Name	Role	Responsibilities
Konstantin Shevtsov	Team Lead / Frontend Dev	Coordination, deadlines, weekly updates, UI structure & UX
Mikhail Makarchuk	Frontend Developer	UI design, CV upload page, results page, usability
Maksim Pinchuk	Backend Developer	API endpoints, CV upload & processing, Supabase integration
Alexandr An	Backend / AI & Testing	AI prompt design, testing, sample CVs, feedback quality

3. System Architecture Overview

The system is a three-tier web application: a Next.js frontend, a Node.js/Next.js backend, and external services (Google Gemini 3.1 Flash and Supabase). All communication happens via REST API over HTTPS.



3.1 Client Layer (Mikhail)

Built with Next.js 14 and Tailwind CSS. Three main pages:

- CV Upload page — drag-and-drop interface; validates file type and size before sending to the API
- Results page — shows the AI score and categorised feedback cards (structure, skills, experience, formatting)
- Auth page — sign-up and sign-in via Supabase Auth

3.2 API Gateway

Next.js API routes act as the entry point. The gateway runs JWT middleware and routes requests to the appropriate backend service.

3.3 Backend Layer (Maksim + Alexandr)

- Auth service — verifies JWT tokens and manages user sessions via Supabase Auth
- File processor — extracts raw text from the uploaded PDF or DOCX file
- AI orchestrator — builds the prompt and calls the Gemini 2.0 Flash API (Alexandr An)
- Result formatter — validates and normalises the Gemini JSON response into a fixed schema
- Storage service — saves the CV file to Supabase Storage and inserts the review record into PostgreSQL

3.4 External Services

Google Gemini 2.0 Flash

Core AI engine. Receives the CV text as a prompt and returns a structured JSON with a score and per-category feedback. Chosen for its speed, low cost, and 1M-token context window. Managed by Alexandr An.

Supabase — PostgreSQL

Stores user accounts, CV upload records, and review results. Managed by Maksim Pinchuk.

Supabase Storage

Stores the original CV files in a secure bucket, linked to the user record by foreign key.

4. Request Workflow

#	Step	Description	Actor
1	Upload CV	User drags a PDF or DOCX file onto the upload page. The frontend validates file type and size (max 5 MB), then sends a POST <code>/api/cv/upload</code> request with the JWT token.	User (Browser)
2	Parse file	The File processor receives the uploaded file and extracts its raw text content. For PDFs the text is read directly; DOCX files are unpacked and their text nodes are joined.	File Processor
3	AI review	The AI orchestrator builds a structured prompt and calls the Google Gemini 3.1 Flash API. The prompt asks Gemini to evaluate the CV across 4 dimensions: structure, skills, experience, formatting. The API returns a JSON object with a score (0–100) and feedback.	Gemini 2.0 Flash
4	Format & save	The Result formatter validates and normalises the Gemini JSON. In parallel, the Storage service saves the CV file to Supabase Storage and inserts the review record into PostgreSQL.	Backend + Supabase
5	Show results	The formatted JSON is returned to the frontend. The Results page renders a score gauge, four feedback cards, and a prioritised list of improvement tips. Past reviews are stored in the user profile.	Results Page

5. Data Model

Core tables in Supabase PostgreSQL:

Table	Key Columns	Description
users	id (UUID), email, created_at	Supabase Auth users
cv_uploads	id, user_id (FK), file_path, uploaded_at	Uploaded CV file references
cv_reviews	id, cv_id (FK), score (INT), feedback (JSONB), created_at	AI review results

6. Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 15, Tailwind CSS	UI, routing, server components
Backend	Node.js, Next.js API routes	REST API, business logic
AI service	Google Gemini 3.1 Flash	CV analysis and feedback generation
Database	Supabase (PostgreSQL)	User data and review results
File storage	Supabase Storage	CV file bucket
Auth	Supabase Auth	JWT-based authentication
Deployment	Vercel / Docker	Hosting and containerisation

7. Key API Endpoints

Method	Endpoint	Description
POST	/api/auth/signup	Register new user
POST	/api/auth/login	Sign in, return JWT
POST	/api/cv/upload	Upload CV file (multipart/form-data)
GET	/api/cv/reviews	List user's past reviews
GET	/api/cv/review/:id	Get single review result